

Experiment- 5

Prism

AIM:

To determine the angle of minimum deviation for a given prism by plotting a graph between the angle of incidence and the angle of deviation.

APPARATUS:

Drawing board, a white sheet of paper, prism, drawing pins, pencil, half-metre scale, office pins, graph paper and a protractor

THEORY:

The angle of minimum deviation is the angle through which a ray of light is deviated from its original path. As the angle of incidence increases, the angle of deviation decreases till it becomes minimum at a particular angle of incidence, and then it increases. The minimum value of the angle of deviation is called the angle of minimum deviation

PROCEDURE:

1. Fix a white sheet on a drawing board using drawing pins.
2. Place a glass prism on it in such a way that it rests on its triangular base. Trace the outline of the prism using a pencil.
3. Draw a thin line NEN normal (perpendicular) to face AB of the prism. Also, draw a straight line PE making an angle preferably between 30^0 and 60^0 as shown in the figure.
4. Fix two pins at a distance of 5cm from each other on the line pE as shown in the figure, later mark these points of the pins as P and Q.
5. Look at the images of the pins, fixed at P and Q, through the other face of the prism, i.e, AC.
6. Fix two more pins, at points R and S vertically, such that the feet of the pins at R and S appear to be on the same straight line as the feet of the images of the pins P and Q when viewed through the face AC of the prism.
7. Remove the pins and the glass prism.
8. Join and produce a line joining R and S; let this line meet the prism at point F.
9. Extend the direction of the incident ray PQE till it meets the face AC. Also extend (backwards) the emergent ray SRF so that these two lines meet at point G.
10. Mark the angle of incidence, $\angle i$, angle of refraction, $\angle r$, and the angle of emergence $\angle e$ and angle of deviation $\angle D$ as shown in the figure.

11. Repeat the experiment for more angles of incidence, preferably between 30° and 60°
12. Plot a graph between the angle of incidence (i) and the angle of deviation (D), and obtain the angle of minimum deviation (D_m) from the graph

RESULT:

- i. The i - D graph indicates that as the angle of incidence (i) increases, the angle of deviation (D) first decreases, attains a minimum value, and then starts increasing for further increases in the angle of incidence.
- ii. Angle of minimum deviation (D_m) =

Precautions

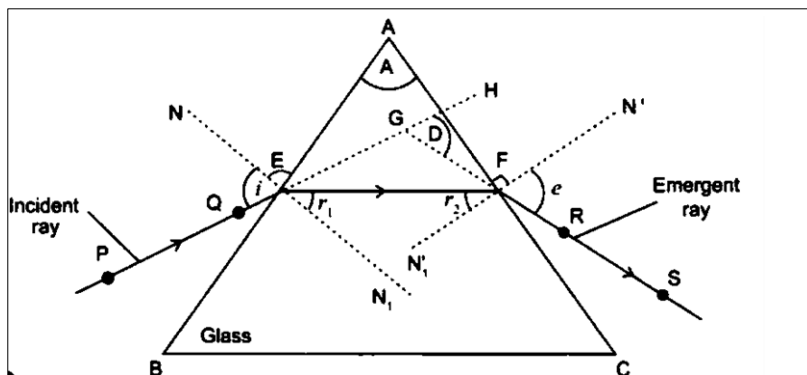
- i. The angle of incidence should be between 35° and 60° .
- ii. The pins should be fixed vertically.

Sources of Error

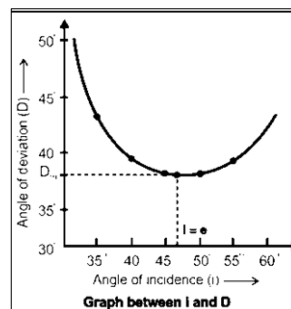
- i. Pin pricks may be thick
- ii. Measurement of angles may be wrong.

Diagram and observation table on the left side of the page

Ray Diagram:



Model graph.



Observation table

S.No	Angle of incidence ($\angle i$)	Angle of deviation($\angle D$)
1		
2		
3		
4		
5		

Calculations:

Angle of minimum deviation =